MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 05 (SURFACE AREAS & VOLUMES)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 13: Surface Areas and Volumes** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show clear steps in calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. The edge of a cube whose volume is equal to that of a cuboid of dimensions 8 cm × 4 cm × 2 cm is: **[1]**

(a) (a) 6 cm (b) 4 cm (c) 2 cm (d) 8 cm

2. Three solid metal cubes of edges 6 cm, 8 cm, and 10 cm are melted to form a single cube. The length of the diagonal of the larger cube formed is: **[1]**

(a) (a) $4\sqrt{6}$ cm (b) $15\sqrt{2}$ cm (c) $12\sqrt{3}$ cm (d) $18\sqrt{3}$ cm

3. Three solid cubes have a face diagonal of $4\sqrt{2}$ cm each. Three other solid cubes have a face diagonal of $8\sqrt{2}$ cm each. All six are melted to form a single cube. The edge length is: **[1]**

(a) (a) 12 cm (b) 24 cm (c) 18 cm (d) 16 cm

4. The sum of length, breadth, and height of a cuboid is $6\sqrt{3}$ cm and diagonal is $2\sqrt{6}$ cm. The total surface area of the cuboid is: **[1]**

(a) (a) 48 cm^2 (b) 72 cm^2 (c) 84 cm^2 (d) 96 cm^2

5. **Assertion (A):** If the volumes of two spheres are in the ratio 343 : 1331, then their radii are in the ratio 7 : 11.
Reason (R): If the radii of two spheres are in the ratio 2 : 3, their surface areas are in the ratio 4 : 9. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. What is the ratio of the volume of a cube to that of the largest sphere that can fit inside it? **[2]**

7. What is the ratio of the volumes of a cylinder, a cone, and a sphere, if each has the same diameter $2r$ and same height $2r$? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. A tent is of the shape of a right circular cylinder up to a height of 3 metres and then becomes a right circular cone with a maximum height of 13.5 metres above the ground. Calculate the cost of painting the inner side of the tent at the rate of 2 per m^2, if the radius of the base is 14 metres. **[3]**

9. A solid consisting of a right circular cone of height 120 cm and radius 60 cm standing on a hemisphere of radius 60 cm is placed upright in a cylinder full of water of radius 60 cm and height 180 cm. Find the volume of water left in the cylinder. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (5 Marks) [5 Marks]

10. A godown building has a vertical cross-section parallel to the width side shaped as a rectangle of size 7 m $\times$ 3 m surmounted by a semi-circle of diameter 7 m. The internal length of the godown is 10 m.

(i) Find the internal volume of the godown building.
(ii) Find the total internal surface area of the building excluding the floor.

[5]

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

11. Case Study: Heritage Architecture – The Great Stupa at Sanchi
The Great Stupa at Sanchi features a grand hemispherical brick dome surmounted by a square cuboidal harmika structure on top. The hemispherical dome has a base radius of 14 metres, while the cuboidal superstructure has dimensions 8 m $\times$ 6 m $\times$ 4 m.

(i) Find the volume of the hemispherical dome of base radius 14 m.
(ii) Find the volume of the cuboidal superstructure.
(iii) Find the curved surface area of the hemispherical dome.
(iv) Find the area of cloth required to cover the exposed hemispherical surface if the cuboidal base (8 m $\times$ 6 m) rests directly on it.

[4]

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— END OF QUESTION PAPER —
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